

A 45-year-old man presented with a 4-day history of fever, headache and neck stiffness. On examination, he was febrile (39°C). His GCS score was 15. Kernig's sign was positive. There were no focal neurological signs. Investigations: CSF: opening pressure= 240 mmH₂O, total protein= 1.2 g/L, glucose= 1.2 mmol/L, cell count 150/μL, lymphocyte count 10%, Gram stain=Gram-positive diplococcus. Which organism is the most likely causative agent?

- A. *Escherichia coli*
- B. *Haemophilus influenzae*
- C. *Listeria monocytogenes*
- D. *Neisseria meningitidis*
- E. *Streptococcus pneumoniae*

ANSWER: E

A 56 year-old man presents for evaluation of tremor. He notes that several family members had similar symptoms in their 50s and 60s. On examination, there is no evidence of hypomimia, bradyphrenia or bradykinesia. His head has a fast frequency tremor. At rest, tremor is not noted. However, with arms raised straight out in front, there is a fairly symmetric, fast, wrist-flexion extension tremor. When writing a spiral, this tremor is even more prominent. Tone is normal. Gait is normal. Which of the following would be considered first line medication to offer in this patient?

- A. Propranolol
- B. Clonazepam
- C. Levodopa
- D. Benzhexol

ANSWER: A

An older lady complains of tingling in her fingers and toes, long-standing lethargy and increasing difficulty walking special during night time due to frequent fall down from imbalance. On examination, her tongue is very red and smooth and there is a sensory loss of vibration and proprioception over the distal limbs with distal muscle strength of 4/5 and hyperreflexia at knee bilaterally. One of the following is not part of the diagnostic work up

- A. CBC
- B. Peripheral morphology
- C. Serum metabolite study
- D. CSF analysis

ANSWER: D

A 15-year-old boy is brought to the emergency department after an accident on his father's farm. The examination reveals weakness of the left lower extremity. There is a loss of pinprick sensation on the right side beginning at the T4 dermatome, and dorsiflexion of the great toe in response to plantar stimulation. What is the name of the syndrome?

- A. Conus medullaris syndrome
- B. Cauda equina syndrome
- C. Brown- Sequard syndrome
- D. Anterior cord syndrome

ANSWER: C

A 7-year-old boy presents with frequent staring episodes. The mother says he pauses during activities, stares off, and then returns to the activity. She can't get his attention during the episodes. He is normal afterward. This occurs multiple times per day. His mother states that he once had an episode while blowing a pinwheel. He is otherwise normal, and his neurologic examination is normal. What is the medication of choice?

- A. Ethosuximide
- B. Lamotrigine
- C. Carbamazepine
- D. Levetiracetam

ANSWER: A

A 39-year-old male with a history of asthma presents to the EOPD with severe SOB for the past 1 day. He also complains of sore throat, malaise, and a nonproductive cough. $T^0 = 37^\circ\text{C}$, HR = 95, RR = 33, BP = 140/82, and $\text{SpO}_2 = 90\%$ on room air. Breathing is labored, and he is speaking in short gasps. Lung auscultation reveals bilateral diffuse expiratory wheezing. What is the appropriate next step in managing this patient?

- A. Cautious oxygen administration targeting $<94\%$
- B. Subcutaneous epinephrine for bronchodilation
- C. Nebulized ipratropium for bronchodilation
- D. Nebulized albuterol for bronchodilation

ANSWER: D

A 61-year-old man presented with 3-month history of progressive shortness of breath and dyspnea on exertion. Since last week he developed lower extremity swelling and shortness of breath at rest. He denies fever, chest pain, or hemoptysis. He has longstanding cough with intermittent yellow phlegm for 8 years but didn't visit a doctor. He is a driver and cigarette smoker. Physical examination is notable for scattered wheezes but no crackles, elevated jugular venous pressure, holosystolic murmur at LLSB and hepatomegaly and grade-II pedal edema. Hemoglobin is 21 g/dl. Which of the following therapies is most likely to prolong his survival?

- A. Enalapril
- B. Diuretics
- C. Oxygen
- D. Metoprolol

ANSWER: C

A 64-year-old woman is admitted to the hospital with a community-acquired pneumonia with a 4-day history of fever, cough, and right-sided pleuritic chest pain. The admission chest x-ray identifies a right lower and middle lobe infiltrate with an associated pleural effusion. All of the following characteristics of the pleural effusion indicate complicated effusion that may require chest tube EXCEPT:

- A. Positive G-stain or culture of the pleural fluid
- B. Lobulated pleural effusion
- C. Pleural fluid pH <7.20
- D. Recurrence of fluid after initial thoracentesis
- E. Pleural fluid glucose <60 mg/dl

ANSWER: D

A 68-year-old man is admitted to the ICU with fevers, hypotension, and hypoxemia. He has felt ill for the past 2 to 3 days with progressive dyspnea at home. On presentation, his room air oxygen saturation is 79%. With a non-rebreather mask, his oxygen saturation remains at 87%. His CXR shows bilateral diffuse coalescent opacities and infiltrates. He is intubated and put on mechanical ventilator. Which ventilator setups should be adjusted to minimize volutrauma?

- A. High tidal volume
- B. Low respiratory rate
- C. Low Tidal volume
- D. High Positive end-expiratory pressure
- E. Low Positive end-expiratory pressure

ANSWER: C

A 57 year old known poorly controlled T2DM patient came to DTCSH with a complaint of left anterior and retrosternal squeezing type of chest pain of 8 hours duration. On arrival at EOPD, he is acutely sick looking, PR=110BPM, BP=140/90mmHg, SPO2=88% on room air. What should be the first investigation to be done for this patient?

- A. Bedside Echocardiography
- B. High sensitivity troponin
- C. 12 lead Electrocardiogram
- D. Chest X- ray

ANSWER: C

A 50 year old male banker came to regular OPD complaining of intermittent occipital headache and easy fatigability for the past 1 year. On P/E, PR=88bpm, BP=170/100, his BMI is 33, ECG showed feature of LVH, serum Cr=1.0 mg /dl. What is the next best step in the management of this patient?

- A. Start antihypertensive drug and advise him on life style modification
- B. No need to start antihypertensive at this time, life style modification is enough
- C. Since it is a single record management is not needed at all
- D. HCT is preferred than ACE/ARBs for this patient

ANSWER: A

The following ECG is an ECG of a 60 years old known COPD patient for the past 6 years who currently presented with palpitation and worsening of dyspnea of 3 months duration. What is the most likely diagnosis based on this ECG?



- A. Paroxysmal supra ventricular tachycardia
- B. Atrial fibrillation
- C. Ventricular tachycardia
- D. Multifocal atrial tachycardia

ANSWER: D

A 35 years old female from Wereta diagnosed to have CRVHD with echo findings of severe MS and moderate TR. All of the following is an expected finding in this patient, EXCEPT?

- A. Shifted apical impulse
- B. Dilated Left Atrium
- C. Wide p wave on ECG
- D. Widening of carina on CXR

ANSWER: A

One of the following is true about HTN?

- A. Secondary HTN occurs in older individuals as compared with essential HTN
- B. Endocrinopathies are by far the commonest cause of secondary HTN
- C. For patients with HTN and proteinuria CKD ARBs are preferred than CCBs
- D. For patients with HTN and ESRD HCT is highly effective anti-hypertensive

ANSWER: C

A 16 years old male high school student came to regular OPD with a complaint of migratory type of multiple joint pain of 5 days duration with associated high grade fever and headache; he claims that he had sore throat 3 weeks back which improved without treatment. On P/E he has swollen and tender bilateral knee and ankle joints. The treating physician planned to investigate the patient. Which of the following is the least likely finding in this patient?

- A. Elevated ESR
- B. MS on echo
- C. MR on echo
- D. prolonged PR interval on ECG

ANSWER: B

A 32 years old mother on her 8th post-partum day after giving Birth through vaginal delivery at health center, presented to a primary hospital with complaints of sudden onset breathlessness, sever chest pain and cough with scanty sputum of 01 day. She also has low grade fever & decreased appetite but she didn't notice any swelling on her lower extremities. On Physical exam, Bp-90/60mmHg, PR-132bpm, RR-38brpm, T°-37.8 °C, PSO2-80%. Which of the following is the appropriate 1st step in the management of the patient condition?

- A. Supplement her with oxygen, hemodynamic support, and/or empiric anticoagulation with heparin.
- B. Start her on broad spectrum antibiotics, oxygen and send her to one of the referral hospitals for chest imaging.
- C. Support with oxygen, start on antibiotics and do lower extremity Doppler study to consider anticoagulation.
- D. Do Chest X-ray, CBC, lower extremity Doppler U/S first and then decide accordingly.

ANSWER: A

45 years old women who was relatively healthy 2 months back, presented with malaise low grade fever, weight loss, and night sweats. She also developed Cough, progressive shortness of breath, mild Chest pain, and lower extremities & abdominal swelling in the past 2 weeks.

Physical examination revealed BP-90/60mmHg, PR-120bpm, and RR-28, T-37.5 ° C, JVP-5 cm from sternal angle, hepatomegaly, ascites, and peripheral edema. Investigations: CBC-Hgb-8mg/dl, ESR-90 mm/hr, Echocardiography shows massive pericardial effusion with no tamponade effect & has pericardial thickening, pericardiocentesis - straw colored fluid, cytology shows lymphocytic effusion. Which of the following indicates the likely Diagnosis and management of the patient's condition?

- A. Malignant pericardial effusion - pericardiocentesis and chemotherapy
- B. Tuberculous pericarditis –do pericardial Biopsy and wait for the results to confirm the Dx.
- C. Tuberculous pericarditis –start on anti TB with or without steroid and continue careful monitoring
- D. Post Myocardial infarction pericarditis – start on MI management
- E. viral or idiopathic pericarditis - start on aspirin or NSAIDs and do Pericardial resection

ANSWER: C

A 29 years old soccer player who was playing his soccer match got sudden short of breath and collapsed in the field. He had no remarkable medical history prior to his incidence. Emergency cardiac imaging was done for him. What findings could be consistent with the patient's condition?

- A. Markedly increased left ventricular wall thickness with decreased dimension and ejection fraction of 70%
- B. Significantly decreased left ventricular wall thickness with marked chamber dilation and ejection fraction of 25%
- C. Increased left ventricular dimension, normal LV wall thickness and normal ejection fraction (55%)
- D. Cardiac tamponade with thickened pericardium

ANSWER: A

All of the following are potential causes of tricuspid regurgitation EXCEPT:

- A. Congenital heart disease
- B. Infective endocarditis
- C. Lateral wall myocardial infarction
- D. Pulmonary arterial hypertension
- E. Rheumatic heart disease.

ANSWER: C

All of the following are correct about skin structures except

- A. Cells take 14 days to reach stratum corenum from basal layer
- B. Fibroblasts, Mast cells and Macrophages are resident cells in dermis
- C. Stratum lucidum is found only on palm and sole
- D. Collagen and elastic fibers are amorphous structures in dermis

ANSWER: D

A patient who visited his family living in a rural area presents to you with an itchy skin lesion involving the whole body including axilla, buttock and groin area of 2 weeks duration. All of the following are correct except;

- A. Scabies is top differential diagnosis
- B. Topical treatment such as 5% Permethrin,BBL are option of treatment
- C. Contact Should be treated if only symptomatic
- D. Fomite treatment is needed

ANSWER: C

A 32-year-old woman visits Diabetes Clinic after uncontrolled glucose level in a trial of 2 weeks' diet and exercise. Her first child was large at birth. She is now 20 weeks' pregnant. She has no prior medical history of note. Her fasting plasma glucose is 120mg/dl and Oral glucose tolerance test at 2 h is 252mg/dl. What is the appropriate treatment option?

- A. Oral glibenclamide
- B. Diet and exercise advice only
- C. Oral metformin
- D. Insulin therapy

ANSWER: D

This is a 55 years old known type II DM female patient on maximum dose of metformin and glimepiride for the past 10 years. Her FBS, post prandial glucose and Hemoglobin A1C levels are in target range for 7 years. Which of the following complications can be best prevented by her strict control?

- A. Stroke
- B. Ischemic heart disease
- C. Retinopathy
- D. Peripheral arterial disease

ANSWER: C

A 30-year-old man is referred for evaluation of refractory hypotension despite 03L of normal saline. He also complains of excessive easy fatigability for the past 01 year. Blood pressure is 60/40 mm Hg and hyper pigmented skin is noticed. Serum sodium is 124 mEq/L (normal 135 to 145mEq/L); potassium, 5.5 mEq/L (normal 3.5 to 5.2 mEq/L). The most likely diagnosis is?

- A. Addison disease
- B. Cushing syndrome
- C. Sipple syndrome
- D. Conn syndrome

ANSWER: A

A 21 year old diabetic patient taking insulin: 6U rapid-acting before each major meal, 30U NPH at bedtime and 12U NPH in the morning. His three day average glucose determination is 120mg/dl before breakfast, 140mg/dl pre-lunch, 200mg/dl presupper and 220mg/dl two hours post dinner. What is the most appropriate measure?

- A. Increase Morning NPH and night Regular insulin
- B. Increasing Morning NPH and increasing Morning Regular insulin
- C. Increasing Night NPH and Increasing Night Regular insulin
- D. Continue same dose NPH and increase night time NPH

ANSWER: A

Which one of the following hormone is not produced from adrenal cortex?

- A. Mineralocorticoids
- B. Androgens
- C. Glucocorticoids
- D. Catecholamines

ANSWER: D

A 70-year-old man is brought to the emergency department with an acute onset profuse vomiting of large amount of bright red blood. He was having mild and intermittent dull type of epigastric pain for the past 3 months. Upper GI endoscopy was done after stabilization and it showed a deep peptic ulcer localized proximally on the lesser curvature of the stomach. This bleeding ulcer is most likely to penetrate which of the following arteries?

- A. Left and right gastroepiploic arteries
- B. Gastroduodenal artery
- C. Left and right gastric arteries
- D. Splenic artery

ANSWER: B

A 35-year-old male presents with substernal discomfort. The symptoms are worse after meals, particularly a heavy evening meal, and are sometimes associated with hot/sour fluid in the back of the throat and nocturnal awakening. The patient denies difficulty swallowing, pain on swallowing, or weight loss. The symptoms have been present for 4 months; the patient has gained weight, 12kg in the past 2 years. What is the best initial approach for this patient is?

- A. Trial of proton pump inhibitors
- B. Barium swallow
- C. Upper GI endoscopy
- D. CT scan of the chest

ANSWER: A

A 20-year-old female presents with a 2-month history of diarrhea with the passage of blood and mucus. Ileocolonoscopy shows inflamed, friable mucosa from rectum to mid-sigmoid. Proximal involvement is not seen. Biopsy reveals inflammation with crypt abscesses. The most likely diagnosis is?

- A. Amoebic colitis
- B. Ischemic colitis
- C. Ulcerative colitis
- D. Tropical sprue

ANSWER: C

In patient with severe hematochezia ,hypotension and anemia both the upper gastroscopy and colonoscopy were not diagnostic .the patient continued to pass clots per rectum .resuscitation has normalized her vital signs and and maintained her HCT at 32%.what is the best next step?

- A. Abdominal CT scan with contrast
- B. Magnetic resonance imaging

- C. Mesenteric angiography
- D. Emergency surgery

ANSWER: C

A 54-year-old man with a history of alcoholic liver disease presents with frank hematemesis, a blood pressure of 80/40 mmHg, pulse rate of 110 and a respiratory rate of 26. After stabilization. What is the best recommended treatment for this patient?

- A. Somatostatin infusion
- B. Balloon tamponed
- C. Endoscopy with variceal ligation
- D. Endoscopy with sclerotherapy

ANSWER: C

A 75-year-old male presented with foul smelling diarrhea for the past 6 months with an average of 4 episodes of oily diarrhea per day. His medical history is notable for hypertension, type 2 DM with complications retinopathy and neuropathy. His appetite is good but he ported 8 kg weight loss within the past 6 months. He has no abdominal pain, blood in the stool, and change in the caliber of the stool. Physical examination is remarkable for an S4 heart sound over the apex and mild stocking glove peripheral neuropathy. Laboratory tests revealed macrocytic anemia and hypo-albuminemia. The most likely diagnosis for this patient is:-

- A. Crohn disease
- B. Chronic pancreatitis
- C. Hemochromatosis
- D. Bacterial overgrowth

ANSWER: D

A 32-year-old woman from fogera wereda presented with 6 month history of fatigue. She reports a normal diet and no gastrointestinal symptoms. Medical history is notable for hypothyroidism and vitiligo. She has no history of drug intake except she is taking levothyroxine since 2 years ago. On physical examination, vital signs are normal. She has several well-defined, symmetrical hypopigmented macules on the extremities and trunk. Laboratory studies: Hgb=8 g/dl (88 g/L), WBC=6000/uL, MCV =112fL, Plt=180,000/uL, Retic count=1% of erythrocytes. Peripheral blood smear is shown macroovalocyte RBC and 2 neutrophils/HPF each having five lobes. What is the most appropriate initial diagnostic test?

- A. Homocysteine level
- B. Methylmalonic acid level
- C. Serum folate level
- D. Serum vitamin B12 level
- E. Bone marrow aspiration and examination

ANSWER: D

An a-29-year old woman Gondar is presented with a complaint of easy bruising of 2 weeks duration. She has no other symptoms, and medical history is unremarkable. She takes no medications. On P/E, vital signs are normal. There is petechiae and small, scattered ecchymosis on the lower extremities. Laboratory studies show a platelet count of 20,000/ul; the remainder of

the CBC is normal. The peripheral blood smear is shown giant platelets which are few in number. What is the most appropriate management?

- A. Glucocorticoids
- B. Plasma exchange
- C. Platelet transfusion
- D. Observation

ANSWER: A

A 25 years old male evaluated at medical OPD for easy fatigability and high grade fever. He has also drenching night sweat and un-quantified significant weight loss. Physical examination is normal except he has sternal bone tenderness. On further investigation he diagnosed to have Hodgkin lymphoma limited to the node of mediastinum measuring 7cm with anterior extension into the sternum. What is the stage of the disease?

- A. II AE
- B. I AE
- C. IV A
- D. IV AE
- E. II BE

ANSWER: E

A -20- year-old male from libokemkem presented with 01 month history of fatigue, low grade fever and anorexia. Medical history is unremarkable and he take no medication. On physical examination vital signs are normal except tachycardia of 110beats/minute. He has pale conjunctivale and palm. Abdominal examination elicited splenomegaly of 6cm along the growth line. Laboratory studies: Hemoglobin =8g/dl, Mean corpuscular volume=850fL, Leukocyte =2,800/ul, Platelet=50,000/ul, rK-39 =negative. Blood film=no hemoparasite seen. Peripheral blood smear showed normocytic normochromic anemia and decreased platelet number. Splenic aspiration didn't reviled any amastigot/LD bodies. What is the best next step of action?

- A. Bone marrow aspiration
- B. Paromomycine and sodium stibogluconate for 17 day
- C. Start suppressive dose of Chloroquine
- D. IV artesunate

ANSWER: A

A 40 -year -old female from Kimir Dingay came for general checkup. She has regular menstrual cycle. V/S -PR-88, BP-110/70mmhg, RR-18 breath/min, T- 36°C, CBC- normal, viral markers -negative RBS-150mg/dl, U/A-leukocyte esterase -+2 blood +3, many WBC, AST -35, ALT-28 Cr-0.9 gm/dl. Which one of the following treatment is indicated?

- A. Ciprofloxacin
- B. IV ceftriaxone
- C. Amoxicillin
- D. No need of treatment

ANSWER: D

A 40-year-old male presented with productive cough of 03 weeks duration. He has low grade intermittent fever, loss of appetite and weight loss (6 kg). Sputum gene xpert-detected, sensitive, PICT- reactive and CD4 count-40. What is the appropriate management?

- A. ART should be initiated first then anti TB
- B. Anti TB then start ART within two weeks
- C. ART and Anti –TB should be initiated simultaneously
- D. Duration of anti TB should be prolonged to 8 months

ANSWER: B

Which one of the following is the recommended test to follow response after eradication therapy of H.pylori infection?

- A. Serology
- B. Histology
- C. Tissue Culture
- D. Urea breath test

ANSWER: D

A 40-year-old known CRVHD patient on monthly B.peniciline injection presented with fever and chest pain of 01 week duration .dental extraction was done 03 weeks back. Echocardiography – 2*3 cm vegetation over mitral valve. The most likely responsible organism is?

- A. Viridian streptococci
- B. Enterococci
- C. Pseudomonas
- D. Klebsella

ANSWER: A

A 55-year-old man known RVI patient on ART for the past 03 years who discontinue the treatment 05 months back after he went to the holy water. Now he presented with pain during swallowing of 02 weeks duration. HEENT-whitish plaque on the buccal mucosa, tongue and oropharynx. What is the recommended 1st line treatment?

- A. Clotrimazole gel
- B. Fluconazole
- C. Itraconazole
- D. Voriconazole

ANSWER: B

Which one of the following types of glomerulonephritis is most characteristically associated with Good pasture's syndrome?

- A. Diffuse proliferative glomerulonephritis
- B. Mesangiocapillary glomerulonephritis
- C. Membranous glomerulonephritis
- D. Rapidly progressive glomerulonephritis
- E. Focal segmental glomerulonephritis

ANSWER: D

You are making round at medical ward and you are reviewing the charts of your patients. Which of the below patients meet the criteria for an acute kidney injury?

- A. Patient with serum K⁺ of 6.8 mmol/l
- B. Patient whose creatinine has increased from 0.9 mg/dl to 1.1 mg/dl over 48 hours
- C. Patient with a creatinine of 1.7mg/dl, whose most recent creatinine one year ago was 1.2mg/dl.
- D. 70kg patient who has produced 20-25 ml of urine per hour for each of the last 8 hours
- E. Patient with an eGFR of 60.

ANSWER: D

Which one of the following is **NOT** a feature of HIV-associated nephropathy?

- A. Normotension
- B. Proteinuria
- C. Elevated urea and creatinine
- D. Focal segmental sclerosis on renal biopsy
- E. Small kidneys

ANSWER: E

A 30-year-old lady presented with polyarthrititis, malar rash and oral ulcerations. Anti-ds-DNA and ANA are present in high titers with normal urinalysis. No other organ involvement. What is the best initial therapy in this lady's management?

- A. Prednisone 1 mg/kg po daily with PPI
- B. Hydroxychloroquine 250 mg po daily
- C. Cyclophosphamide 500mg/m² monthly for 6 months
- D. Azathioprine, 1-2 mg/kg po daily

ANSWER: B

A 71-year-old man has experienced aching pain in the right knee, lower back, right distal fifth finger, and neck over the past 10 years. He notices that the joints feel slightly stiff in the morning, but this passes quickly. The pain is worse toward the end of the day. On physical examination, there is no joint swelling, warmth, or deformity. Some joint crepitus is audible on moving the knee. Laboratory studies show normal levels of serum calcium, phosphorus, alkaline phosphatase, and uric acid. What is the most likely cause of his illness?

- A. Ankylosing spondylitis
- B. Osteoarthritis
- C. Pseudogout
- D. Rheumatoid arthritis

ANSWER: B

A 30 year old man who is HIV positive is starting treatment for Burkitt's lymphoma. His chemotherapy regimen includes cyclophosphamide, vincristine, methotrexate and prednisolone. Around 24 hours after starting chemotherapy, he becomes confused and complains of muscle cramps in his legs. Which one of the following is most likely to have occurred?

- A. Prednisolone- induced psychosis.
- B. Methotrexate pneumonitis leading to hypoxia
- C. Hypercalcemia

- D. Hemorrhagic cystitis leading to acute renal failure
- E. Tumor lysis syndrome

ANSWER: D

A 23 year old boy developed shortness of breath, chest pain and hemoptysis of one hour duration after intramedullary nail was inserted for the indication of right femoral fracture after he sustained road traffic accident a day prior to his admission to orthopedic ward. On physical examination: BP-80/40mmHg, PR-130 bpm and RR-40 bpm and oxygen saturation is 70% at room air. Based on the likely diagnosis, what is the next appropriate investigation?

- A. CT Pulmonary angiography (CTPA)
- B. PA-CXR
- C. Transthoracic echocardiography(TTE)
- D. Lung scan

ANSWER: C

A 16 year old boy presented to the MEOPD with a complaint of coca cola colored urine of 3 days duration. Two weeks prior to the current complaint he was complaining sore throat and was treated with unknown PO medications in a local health center. On P/E: BP=140/90mmHg, has edema and has decreased urine output. One of the following is **LEAST** important next investigations to confirm the underlying cause in these patient?

- A. Sending Throat culture
- B. Measuring ASO titer
- C. Determining Antihyaluronidase antibodies
- D. Sending Renal biopsy

ANSWER: D

A 65 year old smoker complains of Cough with sputum production, and exertional dyspnea of one year duration. On respiratory examination he has prolonged expiratory phase and wheezing. He has barrel shaped chest and uses accessory muscles of respiration. Based on the most likely diagnosis, which one of the below is true?

- A. Steroid will be given if he has previous hospitalization for the same complaint
- B. Steroid will be given if he has previous recurrent pneumonia events
- C. Steroid will be given if he has previous history of mycobacterial infection
- D. Steroids are never indicated for such kind of patients.

ANSWER: A

SURGERY

A 70-year female patient presented with a sudden onset of right upper quadrant pain accompanied by rigor, fever and vomiting. On examination, the blood pressure of 90/50 mmHg, pulse rate of 96 beats per minute and temperature 38⁰c. She had deep icterus of the sclera and right upper quadrant tenderness. What is the most appropriate initial management for this patient?

- A. Fluid resuscitation with crystalloids
- B. Initiate broad spectrum antibiotics
- C. Start him on norepinephrine

D. Immediate cholecystectomy

ANSWER: A

An 18-year-old male patient fall from 5-meter height tree and lands on his left side presented to EOPD with left side chest pain, SOB and left upper quadrant pain. On examination HR is 80 beats/min, and blood pressure is 110/75 mmHg, palpable bony crepitus on left lower 4 ribs and mild LUQ tenderness. Upon investigation, hemoglobin is 13.2 g/dL and CT of the abdomen shows grade III splenic laceration. The patient is admitted to the ward for serial hemoglobin and abdominal monitoring. 8 hours later, the HR is 112 beats/min, blood pressure is 90/50 mmHg, and hemoglobin is 9.1 g/dL. What is the next step in management of this patient?

- A. Exploratory laparoscopy
- B. Exploratory laparotomy
- C. Continue serial examinations and follow up
- D. Transfuse two units of blood, and continue serial examinations and follow up

ANSWER: B

A 45-year-old female patient came to surgical OPD for routine follow up. Three months back she underwent thyroidectomy for toxic multinodular goiter. She claimed that she has excessive sleeping and significant weight gain for the last six weeks. What is the most appropriate next step in this patient care?

- A. Reassure and discharge her from follow up
- B. Start her on levothyroxine
- C. Send thyroid function test(TFT)
- D. Start her on PTU and propranolol

ANSWER: C

A 60-year-old male patient came to surgical OPD with the complaint of dysphagia and significant weight loss of five months duration. He prefers liquid diet than solid food. Barium swallow showed irregularity and shouldering in the lower esophagus. What is the most appropriate next step in this patient care?

- A. Chest CT
- B. Esophagoscopy and biopsy
- C. Surgical resection of the esophagus
- D. Analgesics and discharge because it is advanced disease

ANSWER: B

A 75-year-old man complains of epigastric discomfort. He has noticed his eyes are yellowish for the last two weeks. On evaluation an enlarged gallbladder is palpated. The bilirubin level is 13 mg/dL, the alkaline phosphatase level is 410 U, and the hematocrit is 35%. What is the most likely diagnosis for this patient?

- A. Common hepatic duct tumor
- B. pancreatic head tumor
- C. Cystic duct cholangiocarcinoma
- D. Tumor arising from the ampulla

ANSWER: B

A 55-year-old female patient came to surgical emergency OPD with the complaint of severe abdominal pain of three days duration. The pain was initially peri-umbilical but later involved the whole abdomen. On evaluation, she is acutely sick looking; her blood pressure is, 90/60mmHg, and pulse rate was 110beats/minute, and, respiratory rate was 28breaths/minute, temperature was 39°C. She had distended abdomen and diffuse abdominal tenderness all over the abdomen. What is the most appropriate initial management for this patient?

- A. Fluid resuscitation and adrenaline
- B. Fluid resuscitation and IV antibiotics
- C. Immediately take her to OR and do appendectomy
- D. Send abdomino-pelvic ultrasound and erect chest x-ray

ANSWER: B

A 55-year-old male patient came to a surgical OPD with the complaint of epigastric pain and frequent vomiting which is non-bilious type for one week duration. On evaluation, he is chronically sick looking and emaciated. Succussion splash was positive. Abdominal ultrasound showed thickened gastric antrum and enlarged gastric filled with fluid. What is the most appropriate next step in this patient care?

- A. Barium meal
- B. Abdominal MRI
- C. Send for abdominal CT
- D. Upper GI endoscopy

ANSWER: D

A six-week-old boy was presented with frequent vomiting after he takes breast feed. On evaluation he is pale and skin with turgor. He has ill-defined palpable mass at the right upper quadrant. On abdominal ultrasound, pyloric channel length is 18 mm and pyloric thickness is 6 mm. What is the most appropriate next step in the management of this patient?

- A. Immediate laparotomy
- B. determine serum electrolyte and PH
- C. Send abdominal x-ray
- D. Feeding jejunostomy

ANSWER: B

A 70-year-old male patient presented to the emergency OPD with right side body weakness of 2 days duration with associated failure to control urine. He had no history of trauma. On CT scan it shows 1.5 cm thick hypodense biconcave collection on the right side. What is the most appropriate management for this patient?

- A. Conservatively management with analgesics
- B. Burr holes and drainage
- C. Craniotomy and evacuation
- D. Mannitol

ANSWER: B

A 25-year-old female came to outpatient department because she has pain and swelling over the left knee for the past 4 days. Associated with this she had high grade fever. She is unable to ambulate with limping but walking exacerbates the knee pain. On physical examination, her

temperature was 39.3°C, the left knee is red, warm to touch, tender, and large joint effusion is noted. What is the most appropriate initial step in this patient care?

- A. Arthrocentesis
- B. Arthroscopy
- C. CT scan
- D. MRI

ANSWER: A

A 45-year-old male patient sustained car accident while he was crossing the road. After the trauma he was unable to walk and he had deformed right femur. After admission proximal tibia pin was inserted. He claimed that he become unable to move his right foot after the pin was inserted. What is the most likely explanation for the above complaint?

- A. Femoral nerve injury
- B. Tibial nerve injury
- C. Peronial nerve injury
- D. Sural nerve injury

ANSWER: C

A 45-year-old male patient came to a surgical emergency OPD with the complaint of difficulty of swallowing and fever of three days duration. On evaluation, both adenoid tonsils are enlarged and bulged into oral cavity with whitish exudate. What is the most appropriate initial management for this patient?

- A. Start him on IV medications
- B. Needle aspiration and IV antibiotics
- C. IV antibiotics and tonsillectomy
- D. Start him on PO medications

ANSWER: B

A 70-year-old male patient underwent surgery for squamous cell carcinoma of his nose. What is the best skin substitute for this patient?

- A. Full thickness skin graft
- B. Split thickness skin graft
- C. Composite graft
- D. Healing by second intention

ANSWER: C

A two years old child who was apparently healthy brought by his parents after his mother incidentally detected abdominal mass while bathing his body. Upon physical examination he had vital signs PR=120 RR=24 Temp=36.7 BP=110/80mmHg stage 1 hypertension. Anthropometry evaluation normal. Abdominal exam there is soft non-tender intraabdominal mass which not cross midline of the abdomen. Laboratory exam complete blood count normal, urinary analysis RBC++ other parameters are non-revealing. so what is the most likely diagnosis with its prognosis?

- A. Neuroblastoma with good prognosis
- B. Rhabdomyosarcoma with favorable prognosis
- C. Wilms tumor with favorable prognosis

D. Wilms tumor with poor prognostic features
ANSWER: C

PEDI

A 13 years old girl reports that she has had a fever, weight loss, and night sweats for 3 months. On physical examination, she is found to have painless swelling of the left cervical and supraclavicular lymph nodes. Her liver and spleen are not enlarged. The chest X-ray of the patient reveals mediastinal lymphadenopathy. The best diagnostic test for this case is:

- A. Abdominal CT study
- B. Head CT study
- C. Bone marrow biopsy
- D. Lymph node biopsy

ANSWER: D

A 48 hrs. Old neonate presented with jaundice with a total serum bilirubin concentration of 12 mg/dL. The baby is a 3500-g boy who was born at term to a 27-year-old O-positive, Coombs-test-negative mother. There were no prenatal complications and the mother had regular prenatal care. The baby is breast feeding well with normal vital signs. The most appropriate additional diagnostic studies to evaluate the cause of this infant's jaundice are

- A. Direct serum bilirubin, complete blood count, liver ultrasound, liver transaminases
- B. Direct serum bilirubin, complete blood count, red blood cell glucose-6-phosphate dehydrogenase levels (G6PD), baby blood type
- C. Complete blood count, direct serum bilirubin, baby blood type and Coombs, peripheral smear
- D. Blood cultures, liver transaminases, red blood cell osmotic fragility, direct serum bilirubin

ANSWER: C

During the delivery room resuscitation of a vigorous term newborn, which of the following should be performed first?

- A. Verify the airway is clear, dry, and stimulate the infant.**
- B. The heart rate should be auscultated.
- C. Breath sounds should be auscultated.
- D. The mouth and trachea should be suctioned.

ANSWER: A

A 7 years old boy is admitted to the hospital because of bleeding disorder. Pertinent laboratory findings include a platelet count of 50,000/ μ L, Hematocrit 24%, prolonged prothrombin time, prolonged activated partial thromboplastin time, prolonged thrombin time, and decreased factor VIII level. The most likely cause of his bleeding is?

- A. Immune thrombocytopenic purpura
- B. Vitamin K deficiency
- C. Von willbrand disease
- D. Disseminated intravascular coagulation

ANSWER: D

A 3 year old child comes with generalized body swelling involving genitalia of 3 weeks duration. He had optimal nutritional history and normal anthropometry measurements. Investigations: Complete blood count =normal range, Urine analysis albumin +++, blood =negative, Serum albumin =2.1mg/dl. What is the most likely diagnosis?

- A. Post streptococcal acute glomerulonephritis
- B. Nephritic syndrome
- C. Congenital nephrotic syndrome
- D. Idiopathic nephrotic syndrome

ANSWER: D

A 2yr old patient presented feeding interruption, failure to gain weight and diaphoresis starting 3month of life. He had garde 3 holosystolic murmur best heard at left sternal border. The echocardiographic finding is VSD supracrystal type. The most common complication of patient.

- A. Right ventricular failure
- B. Left ventricular failure
- C. Aortic insufficiency
- D. Arrhythmias

ANSWER: C

A newborn appears with a cyanosis of both hands and feet. The rest of the physical examination is unremarkable. The most likely diagnosis is:

- A. Transposition of great vessels
- B. Gram-negative septic shock
- C. Patent foramen ovale
- D. Unwrapped and cold extremities

ANSWER: D

A child appears in the ER with a shock but adequate reveals warm extremities, tachycardia, bounding peripheral pulses and tachypnea. ABG reveals a mild metabolic acidosis. The most likely diagnosis is:

- A. Hypovolemic shock
- B. Septic shock
- C. Cardiogenic shock
- D. Distributive shock

ANSWER: B

A 9 months old female infant from Gondar town has repeated night awakenings after going to sleep and has been unable to return to sleep without breastfeeding. This behavior has been present for the past month, prior to which she slept the entire night without waking or requiring middle of the night feedings. Her growth and development are normal. She breastfeeds 6-7 times/day and eats a variety of solid foods. Of the following, which is the most likely explanation for this behavior?

- A. Diminishing breast milk production.
- B. Nightmares.
- C. Separation anxiety
- D. Stranger anxiety.

ANSWER: C

GYNI

A 40-year-old P-VI mother who had undergone EVA for GTD 3month back presented with vaginal bleeding of 10hr duration. Upon physical examination there is ~3cm*4cm vascularized easily bleeding mass on periurethral anterior vaginal wall. Serum B-HCG 50,000miu/l. what do you do for her?

- A. Biopsy and histopathology from vaginal mass
- B. Initiate chemotherapy
- C. EVA
- D. TAH + BSO

ANSWER: B

A 25-year-old G-III A-II (6th&5th month of amenorrhea) mother presented with pelvic pressure and increased watery vaginal discharge. Pelvic Ultrasound shows GA=14wk, funneled cervix. What are you going to do for her?

- A. Progesterone supplementation
- B. Appoint for second contact
- C. Insert Pessary
- D. Cerclage

ANSWER: D

A 25-year-old primigravida came for her routine ANC checkup at 39weeks (from 7weeks USG). On physical examination the fundal height is only 32weeks. Her BP is normal and her weight 50kg. You suspect the possibility of IUGR and want to scan with ultrasound. What is the single best parameter for the diagnosis of IUGR?

- A. AC
- B. BPD
- C. FL
- D. Estimated fetal weight

ANSWER: A

A 32-year-old hypertensive pregnant woman at 34weeks comes with complain of continuous abdominal pain, vaginal bleeding and absent of fetal movement for 2 hours. On P/E the SFH corresponds to the period of gestation, uterine contractions are present, FHB is absent. Her PR 80bpm and BP 150/90 mmHg. There is no prior USG report available. What is the next step?

- A. Immediate cesarean section
- B. Admit in labor room for induction of labor
- C. Do PV examination
- D. Send for an urgent USG

ANSWER: D

A 21-year-old primigravida with 37 weeks of gestation present at EOPD complain of headache and blurring of vision since 6 hour. Her BP 170/110 mmHg. Her urine albumin is 3+ by dipstick. The uterus corresponds to 34weeks size and FHB 132bpm. How is the patient managed?

- A. Admit and observe
- B. Admit ,control BP, and continue pregnancy until full term
- C. Admit, start Mgso4 and antihypertensive and terminate the pregnancy
- D. start antihypertensive and follow at OPD

ANSWER: C

A 28-year-old G4P2 mother undergoes a primary cesarean delivery under spinal analgesia for malpresentation of twins at 37 weeks. Immediately after the delivery of the placenta, the anesthetist notes maternal seizure activity with profound hypoxia and hypotension. The patient is intubated and provided with circulatory support with vasopressors. Massive hemorrhage from the surgical site ensues, and the patient is given uterotonic agents and blood products. What is the most likely cause of her hemorrhage?

- A. Amniotic fluid embolism
- B. Halogenated anesthetic agent
- C. Severe preeclampsia with HELLP syndrome
- D. Uterine atony from overdistended uterus

ANSWER: A

A 25-year-old Primigravida at 40 weeks' gestation presents in labor ward with second stage of labor. At the time of delivery, a shoulder dystocia is encountered. A mediolateral episiotomy is performed to assist with dystocia maneuvers. Compared with a midline episiotomy, which of the following is an advantage of mediolateral episiotomy?

- A. Ease of repair
- B. Fewer breakdowns
- C. Lower incidence of dyspareunia
- D. Less chance of extension of the incision

ANSWER: D

W/ro Aster is a 64 year old Para IV Postmenopausal woman for the last 09 years, came to the Gyne OPD with a complaint of Protrusion of mass per vagina of one year Duration. Up on examination, you found that she has 3rd degree UVP with a primary Apical Defect. Which of the following is not a risk for development of her problem?

- A. Multiparity
- B. Anthropoid Pelvis
- C. Previous Instrumental Delivery
- D. Prolonged second stage
- E. Macrosomia

ANSWER: B

Which of the following is the LEAST common outcome of cervical human papillomavirus infection?

- A. Latent infection
- B. Subclinical infection
- C. High-grade dysplasia or cancer
- D. Cervical intraepithelial neoplasia (CIN)

ANSWER: C

Leiomyomas account 1 to 3 percent of infertility cases. Which of the following is **NOT** the putative mechanism by which leiomyomas may disrupt Fertility?

- A. Occlusion of tubal ostia
- B. Disruption of normal uterine contractions
- C. Disruption of implantation secondary to distortion of the endometrial cavity and/or endometrial inflammatory and vascular changes
- D. Disruption of Tubal Motility

ANSWER: D

PUBLIC

There is no single screening test that provides 100% sensitivity and 100% specificity. As a result, it is preferable to apply highly sensitive or highly specific tests depending on the circumstance. Based on this idea, under what circumstances higher sensitive test be preferable than highly specific test?

- A. If the disease has a mild outcome
- B. If the disease is associated with stigma
- C. If the disease is highly infectious
- D. If the disease has a high cost for further investigation

ANSWER: B

If the researcher conducts a study between rural residence and anemia association and gets OR of 5, but the true measure of association was $OR=2$. Which one is true based on the direction of the bias he introduces during the study?

- A. The bias has an effect towards the null
- B. The bias has an effect away from the null
- C. The bias exacerbates the true effect of the factor on the outcome
- D. The bias hides the true effect of the factor on the outcome

ANSWER: C

A laboratory professional working in a certain referral hospital wants to conduct a study to know the occurrence of anemia among children with a history of hook-worm infection and without hook-worm infection and follow for 5 yrs. which study design is appropriate for conducting his study?

- A. Cohort
- B. Case-control
- C. Cross-sectional
- D. experimental

ANSWER: A

A teacher thought that 12% of all students taking Biostatistics course received a grade of 'A'. In a sample of 155 students, he found 21 an 'A'. In this scenario what is the test statistic for testing the true proportion is 12%?

- A. 0.53
- B. 0.01

C. 0.77

D. 0.3

ANSWER: C

A researcher, and he wants to determine the sample interval (represented by k), randomly selecting a number between 1 and k . What is the appropriate sampling technique for the above scenario?

A. Systematic Sampling

B. Simple Random Sampling

C. Stratified Random Sampling

D. Cluster sampling

ANSWER: A

A tablet PC contains 2500 music files. The distribution of file size is highly skewed with many small file sizes. Assume that the standard deviation for this population is 4.5 megabytes (MB). What is the standard error of the average file size when you take a simple random sampling of 36 files from this population?

A. 0.125

B. 0.750

C. 0.014

D. 0.8

ANSWER: B

If health worker says “use contraception to prevent unwanted pregnancy” which one of the following reactions to the health worker’s talk is miss-matched with the type of factors influencing behavior?

A. I understand you but I don’t agree with you –reinforcing factors

B. I would like to use but my husband objects it–reinforcing factors

C. I understand you but I am not interested to use it – predisposing factors

D. Where can I get them- enabling factors.

ANSWER: C

If two project managers Mr.” X” Conduct evaluation to assure the progress of the project and Mr.” Y” did evaluation to see over what was really improved as result of the project after the project was phase out. Which evaluations are describing the works of Mr.”X” and Mr.” Y” respectively?

A. Impact and process evaluations

B. Process and Impact Evaluations

C. Impact and outcome Evaluations

D. Process and outcome Evaluations

ANSWER: D

Suppose there is an assigned Environmental health officer in Gondar town assessed community water adequacy, and his assessment result indicated that each household had one tap at yard level.

A. Basic access

- B. Intermediate access
- C. Optimal access
- D. No access

ANSWER: B

Mrs. A is a 41-year-old female bookkeeper, has been employed by the University of Gondar 12 years ago. She was working poorly at her job, which resulted lower back pain as a consequence of her bad working posture. What is the type of workplace risk in the above scenario?

- A. Psychosocial
- B. Chemical
- C. Ergonomic
- D. Biological

ANSWER: C

Which dietary recommendation do you consider for individuals who lead a sedentary lifestyle and want to control their body weight?

- A. Low nutrient dense foods
- B. High energy dense foods
- C. Low energy dense foods
- D. Low nutrient and high energy dense foods

ANSWER: C

Assuming a 30year man come to you with a case of angina pectoris and you know the patient has private health insurance as a result you prescribe different investigations such as chest x-ray, ECG, echo and blood chemistry. Based on the case which type of health financing problem exists?

- A. Adverse selection
- B. Asymmetric information
- C. Moral hazard
- D. Supplier induced demand

ANSWER: D

Currently our country economic situation is badly affected by war, COVID-19 and drought. Because of this most of the communities expose for different problems and the federal government wants to provide free health service for those segments of the communities. Which type of economic is best match for the scenario?

- A. Positive economics
- B. Micro economics
- C. Normative economics
- D. Macroeconomics

ANSWER: C

A 20year patient come to DTCSH due to acute abdomen and you admit her/him and prescribe different investigation and antibiotics. Based on the case which one is correct.

- A. antibiotics, investigation and bed _ direct medical cost
- B. patient loss his/her work time _direct non-medical cost

- C. pain, stress and tiredness _intangible cost
- D. Antibiotics, investigation and bed---- intangible cost

ANSWER: C

Assuming that University of Gondar, collage of health science wants to achieve its community service in this year by using allocated resource but at the end of the year information's indicated that the service was achieved by using of extra resources. Based on the case which one is correct.

- A. Effective _ efficient
- B. Ineffective _ efficient
- C. Effective _ inefficient
- D. Ineffective _ inefficient

ANSWER: C

A 40 year- old known RVI male patient on TDF/3TC/DTG for the last 2 years present with loss of appetite of 4 month duration. Current CD4 Is 89. Ultrasounds shows matted bowel loops.

What is the most appropriate management for this patient?

- A. Determining the viral load and if it is above 1000, declare treatment failure.
- B. After treating the acute condition, switch to 2nd line ART
- C. Treat the acute condition and enroll him into enhanced adherence support
- D. Treat the acute condition and Hold ART for 2-8 weeks

ANSWER: C

A 72 year old man who is known to have atrial fibrillation come to follow-up clinic. He had transient ischemic attack a week back and takes hydrochlorothiazide for hypertension but is otherwise well. His latest BP is 125/75mmHg.you are discussing on how to reduce his future risk of having stroke. What is his **CHA₂DS₂-VASc** score?

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5
- F. 6

ANSWER: D

